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Mid - Sem
Exam
Set - I
Gr - A

1. Write the correct answer from the options given. (5x1)

(1) Data Science is concerned with:

(i) Handling large volume of data.

(ii) It is an interdisciplinary subject.

(iii) Data science is an important branch of AI.

a) Only (i) is correct. b) (i) and (ii) is correct.

c) (i), (ii), (iii) all are correct. d) None of the above.

2. Data can be of: (i) Quantitative, (ii) Qualitative

(a) Only (i) is correct, (b) only (ii) is correct, (c) Neither (i) nor (ii) is correct. (d) (i) and (ii) both are correct.

3. In univariate data analysis the number of independent variable is _____

(a) One, (b) Two, (c) Many, (d) None of the above.

4. Which is true for Neural Network—

a) Each node computes its weighted input.

b) Node could be in excited state or non-excited state.

c) It has set of nodes and connections.

d) All of the above.

5). In a feed forward neural network with the following specifications -

Input layer has three neurons and one output neuron.

The input values are $[0.3, 0.5, 0.6]$. The weights are $[0.2, 0.1, -0.3]$. The value of the pre-activation to the output neuron is -

a) 1.7 b) -0.07 c) 0.07, d) -1.7

Gr-B

2. Answer the following Question. (5)

if the random variable is discrete in nature then out of PMF and PDF which gives the correct distribution function?

The PMF of a random variable is given by -

$$P(x) = \frac{1}{2^x} \quad \text{if } x \geq 0$$

$$= 0 \quad \text{otherwise.}$$

Calculate $P(x \in A)$ if $A =$ set of odd integers.

Gr-C

3. Answer the following question. (10)

Derive the expression of the delta rule for a single neuron using back ~~prop~~ propagation and gradient descent.